



Bird Species Observation Analysis Report



1. Project Overview

This project analyzes bird species observations across two ecosystems:

- 🌳 Forest
- 🌾 Grassland

The objective is to understand:

- Bird distribution across habitats
- Species diversity and behavior
- Environmental impact on bird activity
- Conservation priorities

This analysis supports **biodiversity conservation, eco-tourism, and environmental planning.**



2. Business Objectives

- Identify **which habitat supports more bird life**
- Understand **species distribution and diversity**
- Analyze **bird behavior patterns**
- Evaluate **environmental impact (weather, time)**
- Identify **at-risk species for conservation**



3. Data Analysis Process



Step 1: Data Collection

- Multiple Excel sheets from forest and grassland datasets

- Each sheet represents a different location
-

◆ **Step 2: Data Preprocessing**

- Combined multiple sheets into one dataset
 - Added:
 - Habitat_Type
 - Admin_Unit
-

◆ **Step 3: Data Cleaning**

- Handled missing values
 - Converted date and time columns
 - Removed duplicates
-

◆ **Step 4: Feature Engineering**

Created new features:

- Year, Month
 - Hour
 - Time_Period (Morning/Afternoon/Evening)
-

◆ **Step 5: Exploratory Data Analysis (EDA)**

Performed:

- Habitat comparison
 - Time analysis
 - Weather impact analysis
 - Species and behavior analysis
-

◆ **Step 6: SQL Analysis**

- Stored data in PostgreSQL

- Executed business queries for insights

◆ Step 7: Visualization

- Built Power BI dashboards:
 - Dashboard 1 → Distribution & Environment
 - Dashboard 2 → Species & Behavior

4. Dashboard Summary

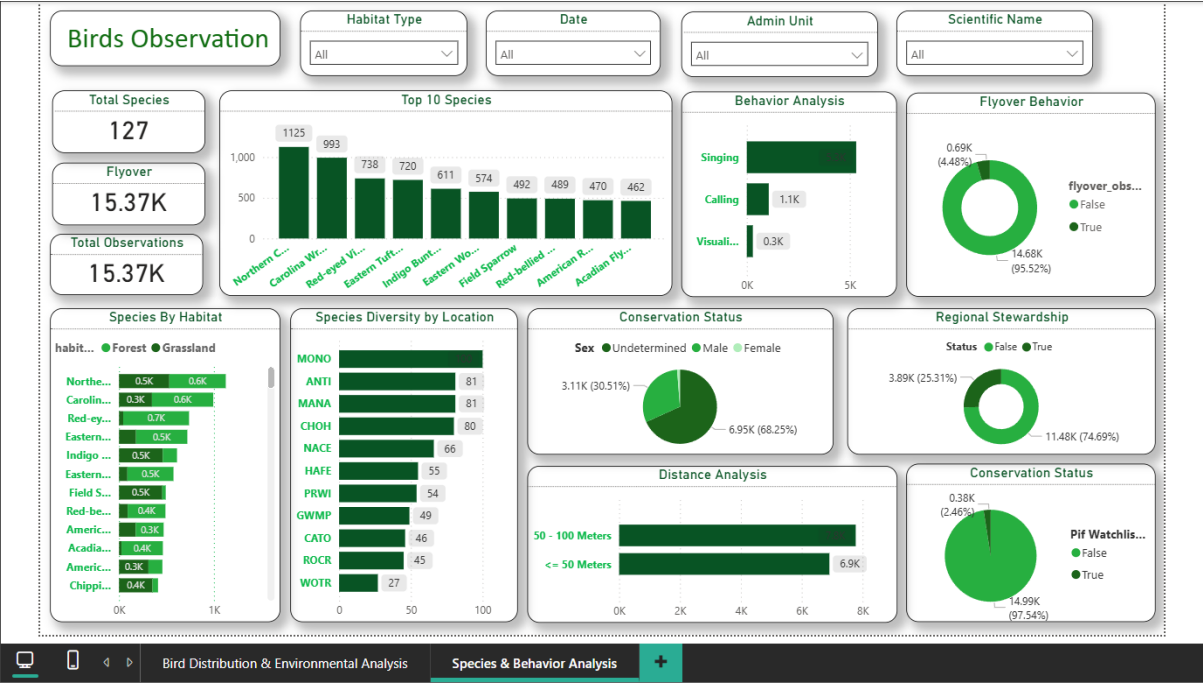
● Dashboard 1: Distribution & Environment

- Habitat comparison
- Location hotspots
- Time trends
- Weather impact



● **Dashboard 2: Species & Behavior**

- Top species
- Behavior patterns
- Distance analysis
- Conservation status



5. Key Insights

🌳 **Habitat Insights**

- Forest has **higher bird population and diversity**
- Grassland supports fewer but specialized species

🕒 **Time Insights**

- Peak activity during **morning**
- Seasonal variation observed across months

Environmental Insights

- Temperature and weather influence bird sightings
 - Clear/moderate weather → higher observations
-

Species Insights

- Few species dominate observations
 - Some species are habitat-specific
-

Behavior Insights

- Most birds identified via **sound (singing/calling)**
 - Limited flyover observations
-

Conservation Insights

- Presence of **watchlist species**
- Some regions require protection

6. Business Questions & Answers

Q1. Which habitat has more birds?

Answer:

Forest habitats show higher bird observations compared to grasslands.

Insight:

Forests provide richer resources (food, shelter), leading to higher bird activity.

Q2. Which habitat has higher species diversity?

Answer:

Forests have a greater number of unique species.

Insight:

Forest ecosystems support more biodiversity due to varied vegetation.

● **Q3. Which locations are biodiversity hotspots?**

Answer:

Certain administrative units (e.g., top-count locations) show significantly higher bird observations.

Insight:

These areas are ideal for:

- Conservation efforts
- Bird-watching tourism

● **Q4. What are the most common bird species?**

Answer:

A few species dominate the observations (Top 10 species).

Insight:

These species are highly adaptable and widespread.

● **Q5. Which species prefer which habitat?**

Answer:

Some species are predominantly found in:

- Forest → dense vegetation species
- Grassland → open-area species

Insight:

Species have strong habitat preferences.

● **Q6. When are birds most active?**

Answer:

Bird activity peaks during:

-  Morning hours

Insight:

Birds are more active during feeding and vocal periods.

● **Q7. How does weather affect bird activity?**

Answer:

- Moderate temperature → higher activity
- Extreme conditions → lower sightings

Insight:

Weather directly influences bird visibility and behavior.

● **Q8. What are common bird behaviors?**

Answer:

Most birds are identified via:

- Singing
- Calling

Insight:

Acoustic detection is the primary observation method.

● **Q9. How far are birds typically observed?**

Answer:

Most birds are observed within shorter distances.

Insight:

Birds tend to stay within localized habitats.

● **Q10. Are there any at-risk species?**

Answer:

Yes, some species fall under the **watchlist category**.

Insight:

These require conservation focus.

7. Business Recommendations

1. Conservation Planning

- Focus on **high biodiversity forest areas**
 - Protect habitats of at-risk species
-

2. Eco-Tourism Development

- Promote bird-watching in hotspot locations
 - Create guided tours for tourists
-

3. Sustainable Land Management

- Maintain ecological balance in grasslands
 - Avoid excessive land conversion
-

4. Environmental Monitoring

- Monitor impact of climate changes
 - Track bird activity over time
-

5. Data-Driven Decision Making

- Use insights for policy-making
 - Support wildlife conservation programs
-

8. Conclusion

This project successfully analyzed bird species data to uncover:

- Habitat-based differences
- Behavioral patterns

- Environmental influences
- Conservation priorities

The insights can be used for:

- Biodiversity conservation
- Sustainable planning
- Eco-tourism growth

9. Future Scope

- Add **Machine Learning models** to predict bird activity
- Include **geographical mapping (GIS)**
- Integrate real-time environmental data
- Build automated monitoring systems

10. One-Line Summary

👉 “This project uses data analysis and visualization to understand bird distribution, behaviour, and environmental impact, helping improve conservation and ecological planning.”

--- Python Code ---

```
import pandas as pd
import matplotlib
import matplotlib.pyplot as plt
import matplotlib.cm as cm
import numpy as np
import seaborn as sns
```

```
forest_file = "E:/Analytics PDF/Bird Species Observation
Analysis/Bird_Monitoring_Data_FOREST.XLSX"
```

```
grass_file = "E:/Analytics PDF/Bird Species Observation  
Analysis/Bird_Monitoring_Data_GRASSLAND.XLSX"
```

```
forest_xls = pd.ExcelFile(forest_file)
```

```
grass_xls = pd.ExcelFile(grass_file)
```

```
print(forest_xls.sheet_names)
```

```
print(grass_xls.sheet_names)
```

```
# Load sheets into dicts --
```

```
forest_data = {}
```

```
grass_data = {}
```

```
for sheet in forest_xls.sheet_names:
```

```
    forest_data[sheet] = pd.read_excel(forest_xls, sheet_name=sheet)
```

```
for sheet in grass_xls.sheet_names:
```

```
    grass_data[sheet] = pd.read_excel(grass_xls, sheet_name=sheet)
```

```
forest_list = []
```

```
grass_list = []
```

```
# Forest
```

```
for name, df in forest_data.items():
```

```
    df['Admin_Unit'] = name
```

```
    df['Habitat_Type'] = 'Forest'
```

```
    forest_list.append(df)
```

```
# Grassland
```

```
for name, df in grass_data.items():
```

```
df['Admin_Unit'] = name
df['Habitat_Type'] = 'Grassland'
grass_list.append(df)
```

```
# Merge all
```

```
forest_df = pd.concat(forest_list, ignore_index=True)
```

```
grass_df = pd.concat(grass_list, ignore_index=True)
```

```
df = pd.concat([forest_df, grass_df], ignore_index=True)
```

```
print(df.shape)
```

```
df.head()
```

```
# Data Cleaning
```

```
# — Preprocessing helper function (applied individually) —————
```

```
def preprocess(df):
```

```
    # Remove duplicates
```

```
    df.drop_duplicates(inplace=True)
```

```
    # Convert date/time
```

```
    df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
```

```
    df['Start_Time'] = pd.to_datetime(df['Start_Time'], errors='coerce')
```

```
    df['End_Time'] = pd.to_datetime(df['End_Time'], errors='coerce')
```

```
    # Extract hour
```

```
    df['Hour'] = df['Start_Time'].dt.hour
```

```
    # Handle missing values
```

```
df['Sex'] = df['Sex'].fillna('Unknown')
df['Flyover_Observed'] = df['Flyover_Observed'].fillna(False)
df['Temperature'] = df['Temperature'].fillna(df['Temperature'].median())
df['Humidity'] = df['Humidity'].fillna(df['Humidity'].median())
```

```
return df
```

```
# — Apply preprocessing individually —————
```

```
forest_df = preprocess(forest_df)
```

```
grass_df = preprocess(grass_df)
```

```
# — Final merge —————
```

```
df = pd.concat([forest_df, grass_df], ignore_index=True)
```

```
print(df.shape)
```

```
df.head()
```

```
# preprocessing after Merging the Datasets --
```

```
'''
```

```
# Remove duplicates
```

```
df.drop_duplicates(inplace=True)
```

```
# Convert date/time
```

```
df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
```

```
df['Start_Time'] = pd.to_datetime(df['Start_Time'], errors='coerce')
```

```
df['End_Time'] = pd.to_datetime(df['End_Time'], errors='coerce')
```

```
# Fix Start_Time format properly
```

```
#df['Start_Time'] = pd.to_datetime(df['Start_Time'], errors='coerce')
```

```

# Extract Hour AFTER conversion
df['Hour'] = df['Start_Time'].dt.hour

# Check if working
print(df['Hour'].isnull().sum())

# Convert time safely
df['Start_Time'] = pd.to_datetime(df['Start_Time'], format='%H:%M:%S', errors='coerce')

# If still null, try general parsing
df['Start_Time'] = df['Start_Time'].fillna(pd.to_datetime(df['Start_Time'], errors='coerce'))

df['Hour'] = df['Start_Time'].dt.hour

# Handle missing values
df['Sex'].fillna('Unknown', inplace=True)
df['Flyover_Observed'].fillna(False, inplace=True)

df['Temperature'].fillna(df['Temperature'].median(), inplace=True)
df['Humidity'].fillna(df['Humidity'].median(), inplace=True)'''

# Post-merge validation --
print(f"Duplicates after merge: {df.duplicated().sum()}")
print(df.isnull().sum())
print(df.dtypes)

# FIX TIME COLUMN
print(df[['Start_Time', 'Hour']].head(10))

# Time Category Analysis (Morning / Afternoon / Evening)

```

```
def get_time_period(hour):
    if pd.isna(hour):
        return "Unknown"
    elif hour < 12:
        return "Morning"
    elif hour < 17:
        return "Afternoon"
    else:
        return "Evening"

df['Time_Period'] = df['Hour'].apply(get_time_period)
```

EDA (Exploratory Data Analysis)

1. Birds Count by Location

```
print(df.groupby('Admin_Unit')['Common_Name'].count().sort_values(ascending=False))
location_counts = df.groupby('Admin_Unit')['Common_Name'].count()
plt.figure()
location_counts.plot(kind='bar')
plt.title("Bird Counts by Location")
plt.xlabel("Location")
plt.ylabel("Count")
plt.show()
```

Birds Count by Location

```
print(df.groupby('Admin_Unit')['Common_Name'].count().sort_values(ascending=False))
location_counts = df.groupby('Admin_Unit')['Common_Name'].count()
plt.figure()
```

```
location_counts.plot(kind='bar')
plt.title("Bird Counts by Location")
plt.xlabel("Location")
plt.ylabel("Count")
plt.show()
```

Forest vs Grassland

```
print(df['Common_Name'].value_counts().head(10))
habitat_counts = df.groupby('Habitat_Type')['Common_Name'].count()
plt.figure()
habitat_counts.plot(kind='bar')
plt.title("Forest vs Grassland Bird Count")
plt.xlabel("Habitat Type")
plt.ylabel("Number of Birds")
plt.show()
```

Birds Count by Year & Month

```
df['Year'] = df['Date'].dt.year
df['Month'] = df['Date'].dt.month

df.groupby('Year')['Common_Name'].count()
df.groupby('Month')['Common_Name'].count()
```

Top 10 Bird Species

```
top_species = df['Common_Name'].value_counts().head(10)

plt.figure()
top_species.plot(kind='bar')
plt.title("Top 10 Bird Species")
```

```
plt.xlabel("Species")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.show()
```

```
# Weather vs Bird Activity
```

```
df.groupby('Temperature')['Common_Name'].count()
df.groupby('Humidity')['Common_Name'].count()
temp_effect = df.groupby('Temperature')['Common_Name'].count()
plt.figure()
temp_effect.plot(kind='bar', figsize = (10,13))
plt.title("Temperature vs Bird Count")
plt.xlabel("Temperature")
plt.ylabel("Count")
plt.show()
```

```
# Time Analysis
```

```
-- # Bird Activity by Hour (Time Analysis)
```

```
df['Hour'] = df['Start_Time'].dt.hour
df.groupby('Hour')['Common_Name'].count()
#df['Hour'] = df['Start_Time'].dt.hour
time_period = df['Time_Period'].value_counts()
plt.figure()
time_period.plot(kind='bar')
plt.title("Bird Activity by Time Period")
plt.xlabel("Time of Day")
plt.xticks(rotation=0)
plt.ylabel("Count")
plt.show()
```

```
hourly_activity = df.groupby('Hour')['Common_Name'].count().dropna()
```

```
plt.figure()  
hourly_activity.plot(kind='line', marker='o')  
plt.title("Bird Activity by Hour")  
plt.xlabel("Hour")  
plt.ylabel("Count")  
plt.show()
```

```
# Observation Duration (Interval)
```

```
interval = df['Interval_Length'].value_counts()  
plt.figure()  
interval.plot(kind='bar')  
plt.title("Observation Interval")  
plt.xlabel("Interval Length")  
plt.ylabel("Count")  
plt.xticks(rotation=0)  
plt.show()
```

```
# Species Analysis
```

```
# Unique species
```

```
df['Scientific_Name'].nunique()
```

```
# Top species
```

```
df['Common_Name'].value_counts().head(10)
```

```
unique_species = df['Scientific_Name'].nunique()
```

```
print("Total Unique Species:", unique_species)
```

#Top Species by Habitat

```
top_species = df['Common_Name'].value_counts().head(10).index
filtered = df[df['Common_Name'].isin(top_species)]
species_habitat = filtered.groupby(['Common_Name', 'Habitat_Type']).size().unstack()
species_habitat.plot(kind='bar')
plt.title("Top Species by Habitat")
plt.xlabel("Species")
plt.ylabel("Count")
plt.xticks(rotation=90)
plt.show()
```

Species Distribution

```
species_dist = df['Common_Name'].value_counts().head(15)
plt.figure()
species_dist.plot(kind='bar')
plt.title("Species Distribution")
plt.xlabel("Species")
plt.ylabel("Count")
plt.xticks(rotation=75)
plt.show()
```

Sex Ratio Analysis

```
sex_ratio = df['Sex'].value_counts()
plt.figure()
sex_ratio.plot(kind='pie', autopct='%1.1f%%')
plt.title("Sex Distribution")
plt.ylabel("")
```

```
plt.show()
```

```
# Behavior Analysis
```

```
df['ID_Method'].value_counts()
df['Flyover_Observed'].value_counts()
id_method = df['ID_Method'].value_counts()
plt.figure()
id_method.plot(kind='bar')
plt.title("Bird Identification Methods")
plt.xlabel("Method")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.show()
```

```
# Flyover Behavior
```

```
flyover = df['Flyover_Observed'].value_counts()
plt.figure()
flyover.plot(kind='pie', autopct='%1.1f%%')
plt.title("Flyover Observations")
plt.ylabel("")
plt.show()
```

```
# Distance Analysis
```

```
distance = df['Distance'].value_counts()
plt.figure()
distance.plot(kind='bar')
plt.title("Observation Distance")
plt.xlabel("Distance")
```

```
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.show()
```

```
# Conservation Analysis
```

```
df['PIF_Watchlist_Status'].value_counts()
df['Regional_Stewardship_Status'].value_counts()
watchlist = df['PIF_Watchlist_Status'].value_counts()
plt.figure()
watchlist.plot(kind='pie', autopct='%1.1f%%')
plt.title("Conservation Status")
plt.ylabel("")
plt.show()
```

```
df.to_csv("cleaned_bird_data.csv", index=False, encoding='utf-8')
```

```
df = pd.read_csv("cleaned_bird_data.csv")
print(len(df.columns))
print(df.columns)
```